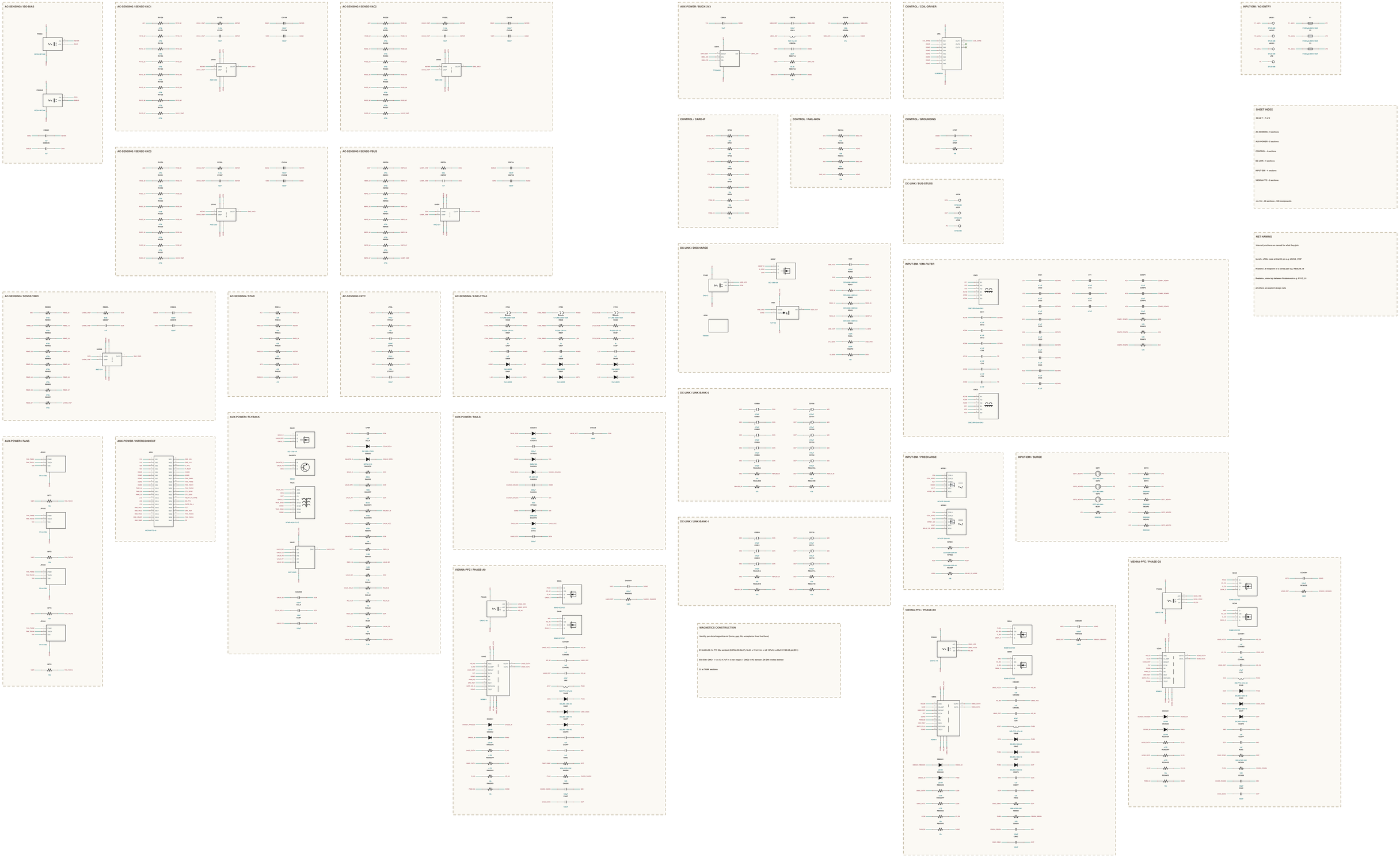
AC-SENSING / ISO-BIAS
AC-SENSING / SENSE-VAC1
AC-SENSING / SENSE-VAC2
AC-SENSING / SENSE-VAC3
AC-SENSING / SENSE-VBUS
AC-SENSING / SENSE-VWD
AC-SENSING / STAR
AC-SENSING / NTC
AC-SENSING / LINE-CTS-0
AUX-POWER / FANS
AUX-POWER / INTERCONNECT
AUX-POWER / FLYBACK
AUX-POWER / RAILS
VIENNA-PFC / PHASE-A0
AUX-POWER / BUCK-3V3
CONTROL / COIL-DRIVER
INPUT-EMI / AC-ENTRY
CONTROL / CARD-IF
CONTROL / RAIL-MON
CONTROL / GROUNDING
DC-LINK / BUS-STUDS
DC-LINK / DISCHARGE
INPUT-EMI / EMI-FILTER
DC-LINK / LINK-BANK-0
INPUT-EMI / PRECHARGE
INPUT-EMI / SURGE
DC-LINK / LINK-BANK-1
VIENNA-PFC / PHASE-B0
VIENNA-PFC / PHASE-C0
SHEET INDEX
SI-UM 7 - 7 of 2
AC-SENSING - 5 sections
AUX-POWER - 5 sections
CONTROL - 4 sections
DC-LINK - 4 sections
INPUT-EMI - 4 sections
VIENNA-PFC - 3 sections
rev D.4 - 29 sections - 335 components
NET NAMING
internal junctions are named for what they join
Uxxxh_vPHx node at that IC pin e.g. UV1h_VINP
Rxxbetw_W midpoint of a series pair e.g. R6bLT4_W
Rxxbetw_xxtmo top between Rxxbetweenhi e.g. RV1D_21
all others are explicit design nets
MAGNETICS CONSTRUCTION
Identity per docs/magnetics.md (turns, gap, Ae, acceptance lines live there)
D1 L&I-LCD: 5x TTR 3ks aerobast (CATALOG.&LST), N=20 w/ 1 tor brim -> L3 107uH, ovrlDuH 0'1/0.5A pk (D31)
EM-EMI: CMC1 = 12x X2 4.7uF in 3 star stages + CMC2 = RC dampac; D6 DM chokes deleted
C1 of 7&NM sections
50 kW ACDC board 1of2 - Vienna PFC (1x air cells)
DC-Modules 50 kW - board ?, sheet ? of 2
2026-09-06 · rev D.4